

Extrema-Frame Foreground-Background Theory: Self-Dual Structure, Exact Common-Realization Semantics, and Ownership Complexity

Alp Eren Bütün

Independent Researcher, Türkiye

ORCID: 0009-0001-2479-6037

github.com/alperenbutun

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Abstract

We develop **Extrema-Frame Foreground-Background theory (EF-FG)**, a self-dual mathematical framework for foreground/background structure in finite scalar fields. EF-FG separates a **spatial-frame projection**, which records competitor/carrier geometry around extrema, from a **polarity-core ownership projection**, which classifies an observer point as bright-owned, dark-owned, background, or ambiguous. For the ownership layer, the full elder hierarchy reduces to an **observer-record chain**: nonrecords are eliminable, and record death is recovered exactly from the running elder attachment at the observer.

For interval-valued scalar fields we impose **exact common-realization semantics**: births, attachments, elder order, record identities, and ownership must arise from one scalar realization rather than from independently optimized marginal envelopes. This yields two complementary complexity phenomena. Local possible-exceedance is polynomial-time decidable on interval-valued cactus graphs, whereas global same-realization safety is NP-complete already on radius-two trees. We isolate a two-coordinate **q-band owner-scheduling kernel**, prove its NP-completeness, lift it exactly to interval-valued radius-two trees, and obtain NP-completeness of **Possible Background Polarity-Core Ownership** for fixed $\theta = 1/2$ and $\Delta = 0$. Separately, common-realization obstruction order is unbounded: for every $r \geq 1$ we construct protected strict-record families whose feasibility complex is the simplex boundary, with canonical and section-free radius-two realizations. The contribution is the EF-FG semantics and this theorem package, not the underlying classical notions of merge trees, self-dual morphology, or interval uncertainty.

Keywords. scalar fields; foreground/background; persistence; merge trees; uncertainty; common realization; computational complexity; NP-completeness; local-to-global obstruction.

1 Introduction

Foreground/background separation is usually presented as a labeling problem. EF-FG instead asks a structural question: **what scalar-field structure should count as foreground or background, and how should that structure behave under contrast inversion, topological ambiguity, and bounded perturbation?**

Let $G = (V, E)$ be a finite connected graph and $F : V \rightarrow \mathbb{R}$ a scalar field. Bright structure is generated from superlevel extrema; dark structure is obtained from the same construction applied to $-F$. The theory is built around four requirements.

1. **Extremal grounding.** Foreground structure is tied to extrema, persistence, and maximin connectivity rather than to an arbitrary partition.

2. **Self-duality.** Replacing F by $-F$ exchanges bright and dark roles without changing the form of the theory.
3. **Exact uncertainty.** Under interval uncertainty, all quantities used in one decision must come from one common scalar realization.
4. **Explicit algorithmic boundary.** The theory distinguishes tractable local-witness questions from globally coupled common-realization questions.

EF-FG has two outputs that must not be conflated. The **spatial-frame projection** records competitor sectors, carriers, terminal regions, and frame geometry. The **polarity-core ownership projection** records whether an observer x is owned by a significant bright structure, by a significant dark structure, is background, or is bright/dark ambiguous. We write

$$\boxed{\text{SF}_x \neq \text{PO}_x}$$

as a scope principle: the two projections interact, but neither determines the other in general.

The starting topological object for one polarity is the merge geometry of the scalar field. If p is a birth atom, B_p its birth value, and x an observer, define the maximin attachment

$$\kappa_F(p, x) = \max_{\pi: p \rightsquigarrow x} \min_{v \in \pi} F(v).$$

A direct implementation could carry the full merge hierarchy. One of the central structural results below shows that pointwise ownership does not need it: only **observer records**, i.e. births that strictly raise the running elder attachment at x , can contribute positive observer-core saliency.

The uncertainty model is an interval box

$$\mathcal{P} = \prod_{v \in V} [\ell_v, u_v].$$

The key semantic choice is relational rather than marginal. A tuple of birth levels, attachments, elder pressures, record identities, and ownership labels is admissible only when one $F \in \mathcal{P}$ realizes the entire tuple simultaneously. This prevents a common error: combining coordinate-wise extrema that are individually attainable but mutually incompatible.

This common-realization requirement is the source of the main complexity boundary. A local query such as

$$\exists F, \rho : \Sigma_\theta^g(x; F, \rho) > H$$

can admit a bounded witness formulation. By contrast, a global safety query asks for one realization that suppresses all dangerous records simultaneously. We show that this synchronization problem is NP-complete even on radius-two trees.

The proof proceeds through three layers. First, the observer inequalities are conjugated to a low-dimensional **q-band owner-scheduling kernel** with state (h, x) . Second, this kernel is realized exactly by interval-valued grayscale branches $x - s_i - p_i$. Third, a constant-size dark anchor neutralizes the opposite polarity and lifts one-polarity safety to self-dual possible-background ownership.

A separate theorem family shows that the difficulty is not only computational. For every $r \geq 1$ there are $r + 1$ protected strict-record obligations for which every proper subfamily has one common realization but the full family has none. Hence no bounded local consistency order can certify global common realizability in general.

1.1 Main contributions

The paper establishes the following package.

- A self-dual EF-FG model with separate spatial-frame and polarity-core projections.
- A hierarchy-free observer-record normal form for pointwise persistent-core saliency.
- Exact common-realization semantics for interval uncertainty.
- An exact q-band transition law, an $O(n^2)$ headroom-indexed dynamic program for arbitrary fixed event orders, and a linear-time greedy specialization for nonincreasing-headroom orders.
- NP-completeness of q-band owner scheduling for fixed $\theta = 1/2$ and $H = 0$.
- An exact physical lift to interval-valued radius-two trees.
- NP-completeness of one-polarity common-realization observer safety on those trees.
- NP-completeness of Possible Background Polarity-Core Ownership at fixed $\theta = 1/2$, $\Delta = 0$.
- Canonical and raw simplex-boundary realization theorems $K_\sigma(P_r) = \partial\Delta^r$ and $K_{\text{raw}}(P_r) = \partial\Delta^r$ for all $r \geq 1$.
- A reference implementation architecture and executable regression suite separating deterministic evaluation from certified interval queries.

The article is **self-contained**. Technical proofs and construction details are collected in Appendices A–G of the same document; executable regression logs remain reproducibility artifacts rather than proof premises.

1.2 Relation to prior work

Topological merge trees are established scalar-field summaries, with both structural and scalable computational treatments [14, 13]. Separately, hierarchical region-merge trees have long been used in image segmentation [2]. Self-dual mathematical morphology is likewise classical: the tree of shapes gives a contrast-invariant hierarchy of upper- and lower-level-set structure [1], and extremum significance has long been studied through dynamics/extinction values on component trees [3]. Later work supplied an n -dimensional proof of the tree-of-shapes construction [6], and recent work develops consistent connected operators directly on trees of shapes [7]. A 2026 preprint also uses interval-valued maps as a computational/topological device for tree-of-shapes construction [8]; that use of interval values is distinct from the interval box here, which represents uncertainty over admissible scalar realizations.

Uncertain scalar-field topology has been studied through mandatory critical points and mandatory join/split information [4], structural averages of merge trees [5], statistical summaries of Morse-complex ensembles [12], and probabilistic critical-point uncertainty [11]. Hardness is also known for topological reconstruction and simplification problems [9].

Accordingly, EF-FG does **not** claim novelty for extrema, persistence, self-duality, interval uncertainty, maximin paths, or topological hardness individually. Its claims are narrower: observer-relative foreground ownership under exact same-realization semantics, the observer-record normal form, the local/global complexity separation on shallow graph classes, and explicit all-orders common-realization obstruction families. These statements are theorem-level properties of the EF-FG semantics rather than priority claims about the neighboring ingredients.

2 EF-FG model

2.1 Filtration conventions and birth atoms

For $t \in \mathbb{R}$, let $G^{\geq t}$ be the induced subgraph on $\{v : F(v) \geq t\}$. A **birth plateau** at level B is a connected local-maximum plateau $P \subseteq F^{-1}(B)$ whose superlevel component appears when the threshold reaches B . The corresponding component identity is a **primitive birth atom** λ , with birth level $B_\lambda = B$. A **birth-carrier representative** is any vertex of P ; all such representatives have the same maximin attachment to any observer because they are connected inside $G^{\geq B}$.

As the threshold decreases, birth components merge. The elder rule keeps the atom of larger birth level. If several atoms have equal birth level, an **admissible elder/tie resolution** is a total order on those equal-birth atoms, used consistently at later merges. Exactly one atom survives to the bottom of the connected filtration; it is the **essential atom**. Every other atom is **finite** (or **nonessential**), and its **elder death level** $d(\lambda)$ is the merge level at which it is killed by an older atom. The dark filtration is the same construction applied to $-F$.

2.2 Scalar fields, births, and maximin attachment

Fix a finite connected graph $G = (V, E)$ and a scalar field $F : V \rightarrow \mathbb{R}$. For the bright polarity, let λ range over primitive nonessential birth atoms of the superlevel filtration. Write

$$B_\lambda = \text{birth level of } \lambda, \quad d(\lambda) = \text{elder death level.}$$

For an observer $x \in V$, define

$$a_\lambda(x) = \kappa_F(\lambda, x),$$

where κ_F is maximin connectivity. If $O(\lambda)$ denotes the elder atoms of λ , define the observer elder pressure

$$D_x(\lambda) = \max_{\mu \in O(\lambda)} a_\mu(x).$$

The dark polarity is the bright construction applied to $-F$.

2.3 Observer records

A nonessential birth λ is an **observer record at** x if

$$a_\lambda(x) > D_x(\lambda).$$

Thus records are precisely the births that strictly raise the running elder attachment available at the observer.

A basic maximin identity drives the theory.

Lemma 2.1 (maximin equality). *If*

$$\kappa_F(p, x) > \kappa_F(q, x),$$

then

$$\boxed{\kappa_F(p, q) = \kappa_F(q, x).}$$

Proof sketch. The strong-triangle inequality gives the lower bound $\kappa_F(p, q) \geq \kappa_F(q, x)$. If the left side were larger, concatenating a maximizing p - q path with a q - x path would force q to attach to x above $\kappa_F(q, x)$, a contradiction. A complete proof is given in Appendix A, Section A.1. $\square$

2.4 Observer-core saliency

Fix $0 < \theta < 1$. A finite birth λ contributes to the observer core when

$$a_\lambda(x) > \theta B_\lambda + (1 - \theta)D_x(\lambda).$$

The one-polarity observer saliency is

$$\Sigma_\theta(x; F) = \max\left(\{B_\lambda - D_x(\lambda) : a_\lambda(x) > \theta B_\lambda + (1 - \theta)D_x(\lambda)\} \cup \{0\}\right).$$

Set

$$\Sigma_\theta^+(x; F) = \Sigma_\theta(x; F), \quad \Sigma_\theta^-(x; F) = \Sigma_\theta(x; -F).$$

For a significance threshold $\Delta \geq 0$, the polarity-core ownership label is

$$\text{Own}_{\Delta, \theta}(x; F) \in \{\text{bright, dark, background, ambiguous}\},$$

with background defined by

$$\max\{\Sigma_\theta^+(x; F), \Sigma_\theta^-(x; F)\} \leq \Delta.$$

The remaining labels are determined by which polarity exceeds Δ , with an explicit ambiguous state when both do.

2.5 Spatial-frame projection

Fix a spatial seed threshold $\Delta_{\text{seed}} \geq 0$. For one polarity, the selected seed family consists of the finite birth atoms with persistence $B_\lambda - d(\lambda) > \Delta_{\text{seed}}$; the essential atom is excluded. Choose one birth-carrier representative for each selected atom p and put

$$\kappa_p(v) = \kappa_F(p, v), \quad \beta(v) = \max_r \kappa_r(v), \quad W(v) = \{p : \kappa_p(v) = \beta(v)\}.$$

The **rooted co-winner basin** $\mathcal{B}_p$ is the connected component containing p in $\{v : p \in W(v)\}$. Two selected seeds p, q are **carrier-adjacent** when $\mathcal{B}_p \cap \mathcal{B}_q \neq \emptyset$. Their gate scalar is

$$s(p, q) = \max_{v \in \mathcal{B}_p \cap \mathcal{B}_q} \beta(v),$$

and the connected components attaining this maximum are the top gate components. Route incidence through an observer, midpoint margins, designated boundary terminals, and correlated residual terminal regions then form the spatial-frame state; their exact definitions are given in Appendix B.

Scope principle. Polarity-core ownership is an observer/persistence projection, whereas spatial frame is a carrier/sector projection. The two are evaluated from the same scalar realization, but neither determines the other in general. The full fixed-realization EF-FG state stores both rather than identifying them.

2.6 Interval uncertainty and exact semantics

An interval-valued scalar field is

$$\mathcal{P} = \prod_{v \in V} [\ell_v, u_v].$$

A **realization** consists of one $F \in \mathcal{P}$ together with an admissible resolution of ties/elder order. A property is **possible** if one realization satisfies it and **robust** if every realization satisfies it.

This quantifier discipline is essential. If $A(\omega)$ and $B(\omega)$ are observables on the realization space, then generally

$$\{(A(\omega), B(\omega)) : \omega \in \Omega\} \subsetneq \{A(\omega) : \omega \in \Omega\} \times \{B(\omega) : \omega \in \Omega\}.$$

Thus one must **join before projecting**: a possible ownership label cannot in general be inferred by combining independently attainable extrema of births, attachments, or saliencies.

3 Hierarchy-free observer normal form

The full merge hierarchy can be eliminated from pointwise observer-core evaluation.

3.1 Exact observer-record death

Theorem 3.1 (exact observer-record death). *Let λ be a nonessential primitive birth atom. If*

$$a_\lambda(x) > D_x(\lambda),$$

then λ is an observer record and

$$\boxed{d(\lambda) = D_x(\lambda)}.$$

If instead $a_\lambda(x) \leq D_x(\lambda)$, then λ cannot make a positive observer-core contribution.

Proof sketch. For a record, choose the elder atom attaining $D_x(\lambda)$. Lemma 2.1 identifies the merge level between the younger record and that elder atom with the elder's attachment at x , so the ordinary elder death is exactly $D_x(\lambda)$. If λ is not a record, its attachment does not exceed elder pressure and therefore cannot satisfy the strict core inequality. $\square$

A complete proof is given in Appendix A.

An immediate consequence is the formula already used in Section 2.3: the saliency maximum can be taken over all births because nonrecords eliminate themselves automatically.

3.2 Record chains

Order the finite observer records by elder-to-younger birth order,

$$r_1, r_2, \dots, r_m,$$

and write $a_i = a_{r_i}(x)$. Include the essential elder supply as a_0 . Then

$$\boxed{d(r_i) = a_{i-1}}.$$

Thus the complete finite record history is a chain of adjacent transitions. If B_i is the birth of r_i , then the i -th potential contribution is determined by the triple

$$(B_i, a_{i-1}, a_i).$$

For an H -safety query, a record is dangerous precisely when

$$B_i - a_{i-1} > H$$

and

$$a_i > \theta B_i + (1 - \theta)a_{i-1}.$$

Equivalently, define the required predecessor supply

$$L_i(H) = \min \left\{ B_i - H, \frac{a_i - \theta B_i}{1 - \theta} \right\}$$

with the obvious convention that only the active branch matters. Then r_i is H -safe iff a_{i-1} exceeds the corresponding threshold. This is the local origin of the later scheduling formulation.

3.3 A single elder-supply profile

The record chain may be represented by one rooted supply profile $E_x(b)$: the greatest observer attachment supplied by births older than threshold b . Its jumps are exactly the observer records. At a finite jump (B, D, A) ,

$$D = \text{exact elder pressure}, \quad A = \text{record attachment}, \quad B - D = \text{exact persistence}.$$

Hence

$$\boxed{\Sigma_\theta(x; F) \leq H \iff \text{every finite jump } (B, D, A) \text{ satisfies } B - D \leq H \text{ or } A \leq \theta B + (1 - \theta)D.}$$

On rooted trees the profile obeys a max/min recurrence over child subtrees, giving a direct deterministic evaluation algorithm. A proof of the record-chain reduction and the rooted supply recurrence is given in Appendix A.

3.4 Self-duality

All statements above are polarity-normalized. Applying them to $-F$ gives the dark theory with no change of form. In particular, grayscale inversion transports the entire common-realization relation bijectively: bright-possible becomes dark-possible, bright-robust becomes dark-robust, and the full ownership relation is conjugated realization by realization.

4 Exact common-realization perturbation theory

4.1 Why marginal envelopes are insufficient

Suppose two admissible realizations ω_0, ω_1 give

$$(A(\omega_0), B(\omega_0)) = (0, 1), \quad (A(\omega_1), B(\omega_1)) = (1, 0).$$

Then each coordinate individually attains both 0 and 1, but $(0, 0)$ and $(1, 1)$ are unrealizable. This elementary observation becomes structural in EF-FG because births, attachments, elder pressure, record identity, and ownership are coupled by the same field values.

We therefore define the **exact EF-FG realization relation** by evaluating all requested coordinates on one realization first and projecting only afterward. Possible and robust states are existential and universal statements over that relation.

4.2 Ownership relation

For a fixed observer x , a realization determines

$$(\Sigma_\theta^+(x), \Sigma_\theta^-(x), \text{Own}_{\Delta, \theta}(x), \text{SF}_x, \text{record data}).$$

The exact perturbation object is the image of the realization space under this joint map. In particular, a claim that x is possibly background means

$$\exists F, \rho : \Sigma_{\theta}^{+}(x; F, \rho) \leq \Delta \quad \text{and} \quad \Sigma_{\theta}^{-}(x; F, \rho) \leq \Delta$$

for the **same** F and resolution ρ .

4.3 Realization-wise self-dual conjugacy

Theorem 4.1 (realization-wise self-duality). *Grayscale inversion $F \mapsto -F$ induces a bijection between the bright exact relation over $\mathcal{P}$ and the dark exact relation over $-\mathcal{P}$. Consequently possibility, robustness, background status, and deterministic bright/dark ambiguity are preserved under polarity exchange in the expected self-dual sense.*

The theorem follows because inversion is a bijection on realizations and every dark construction is defined as the bright construction on $-F$.

Proof details are included in Appendix A.

4.4 Local witness versus global synchronization

The exact relation already suggests two algorithmic regimes. A local exceedance needs one feature witness and a bounded set of separator constraints. A global safety condition quantifies over the entire realized record history. Later sections make this distinction exact: the former is polynomial on interval-valued cacti, while the latter is NP-complete already on radius-two trees.

5 The q-band owner-scheduling kernel

We now isolate the arithmetic core of one-polarity common-realization safety.

5.1 Hard-band state

Write the running state as (M, D) , where M is the current birth cap and D is the current elder supply. For one event i , let its birth interval be $[\ell_i, u_i]$ and its attachment interval be $[\alpha_i, \beta_i]$. In the hard band, safety is

$$a \leq \theta B + (1 - \theta)D.$$

Fix a critical birth floor q . Events with $\ell_i = q$ must be resolved before the cap drops below q . When $q \leq D + H$, the band is easy: an H -safe suffix is immediately reachable. The nontrivial regime is therefore

$$q > D + H.$$

For a hard-band event define the maximal safe choice

$$U_i = \min\{M, u_i\}, \quad a_i^* = \min\{\beta_i, \theta U_i + (1 - \theta)D\}.$$

Every other q -preserving safe realization of the event is dominated by this choice.

5.2 q-band coordinates

Set

$$h = M - q, \quad x = q - D,$$

and for event i define

$$h_i = u_i - q, \quad A_i = q - \alpha_i, \quad B_i = q - \beta_i.$$

For general θ let $c = 1 - \theta$ and $s_i = \min\{h, h_i\}$. Then an owner use is legal exactly when

$$\boxed{B_i < x}$$

and

$$\boxed{cx - \theta s_i \leq A_i},$$

with deterministic transition

$$\boxed{(h, x) \mapsto (s_i, \max\{B_i, cx - \theta s_i\})}.$$

For the hardness specialization $\theta = 1/2$ this becomes

$$(h, x) \mapsto \left(s_i, \max \left\{ B_i, \frac{x - s_i}{2} \right\} \right),$$

legal when

$$B_i < x, \quad \frac{x - s_i}{2} \leq A_i.$$

5.3 Fixed-order reachability

The scalar transition is monotone for a fixed effective headroom, but a used event can lower the future headroom. Consequently one scalar minimum does not suffice for an arbitrary fixed order. The correct state retains the current headroom as well.

Theorem 5.1 (fixed-order absorbing-tail reachability). *For any fixed event order, whether some admissible subsequence reaches the absorbing safe tail $x \leq H$ is decidable by a headroom-indexed dynamic program using $O(n^2)$ event-state transitions. If additionally*

$$h_0 \geq h_1 \geq \dots \geq h_n,$$

then every used event has effective headroom h_i , independent of previous skips, and the one-pass prefix-minimum greedy is exact in $O(n)$ transitions.

A proposed owner word is still verified in $O(n)$ exact transitions, so NP membership does not rely on the stronger greedy special case.

Remark 5.2 (why the headroom hypothesis matters). At $\theta = 1/2$, $H = 0$, start from $(h, x) = (10, 10)$ and fix the order

$$e_1 = (0, 10, 0), \quad e_2 = (10, 0, 0).$$

Using e_1 first gives $(0, 5)$, after which e_2 is infeasible because $(5 - 0)/2 > 0$. Skipping e_1 and using e_2 gives $(10, 0)$ and reaches the safe tail. Hence a single prefix minimum in x is not valid without retaining headroom information.

The dynamic program and the nonincreasing-headroom greedy proof are given in Appendix C.

6 NP-completeness of q-band owner scheduling

6.1 Decision problem

The q-band owner-scheduling completion problem asks whether a finite event family admits an order and legal owner-use pattern that reaches the absorbing safe tail. We use the fixed specialization

$$\theta = \frac{1}{2}, \quad H = 0.$$

Membership in NP follows from a rational schedule certificate: one checks the event order and the deterministic q-band transitions in polynomial time.

6.2 Reduction spine

NP-hardness is by reduction from Monotone NAE-3-Sat-E4 [10]. The main logical spine is as follows; the complete reduction, including the binding scale and phase-leak repairs, is given in Appendix D.

Each variable module has exactly two successful traversal modes. The choice designates one of two payload banks for deletion and leaves the other available for clause auditing. A two-gate **headroom certificate** prevents a payload token from being replaced by a token belonging to another variable or another truth bank. Clause modules use a three-token not-all-high audit, together with directed self-policing, so that a clause succeeds exactly when its three incoming truth states are not all equal. A homogeneous scale hierarchy isolates variables from one another and global audit floors prevent cross-module substitutions. A final finite control wrapper closes premature phase entry and early-pivot leakage.

Completeness maps a satisfying NAE assignment to the corresponding branch of each variable module and then executes all clause/audit modules. Soundness reconstructs a Boolean choice from every successful schedule and uses the headroom certificates plus the self-policing clause gadgets to show that every clause is NAE-satisfied.

Theorem 6.1 (q-band NP-completeness). *Q-band owner-scheduling completion is NP-complete even for*

$$\theta = \frac{1}{2}, \quad H = 0,$$

with rational dyadic event parameters and a linear number of events.

The result isolates the source of hardness: **global order selection under one-use owner resources**, not evaluation of a fixed order.

The complete iff reduction and NP-membership proof are in Appendix D.

7 Exact lift to interval-valued scalar trees

The scheduling kernel is not merely an abstract surrogate. It is exactly realized by radius-two scalar graphs.

7.1 Affine rectangle conjugacy

Choose q larger than all q-band coordinates and apply

$$M = q + h, \quad D = q - x,$$

and, for every event,

$$\ell_i = q, \quad u_i = q + h_i, \quad \alpha_i = q - A_i, \quad \beta_i = q - B_i.$$

Under this change of variables, the maximal-safe rectangle transition is conjugate to the q -band transition. Before the physical tree is built we apply the positive-floor normalization proved in Appendix E: every convenience floor $B_i = 0$ is replaced by one common rational $\varepsilon > 0$ chosen below every intended positive transition scale. Appendix E proves by a dominance coupling that this normalization preserves YES/NO equivalence; in particular it neither destroys the intended completeness schedules nor creates a false YES instance. After normalization,

$$0 < \varepsilon \leq B_i \leq A_i$$

for every event.

7.2 Radius-two physical realization

For each event create a branch

$$x - s_i - p_i,$$

where p_i is an interval-valued peak and s_i an interval-valued saddle. After positive-floor normalization,

$$F(s_i) \leq q - \varepsilon = F(x) < q \leq F(p_i).$$

Since the branch path is unique and the observer is no lower than the saddle,

$$\kappa_F(p_i, x) = F(s_i).$$

An initial essential branch sets the starting elder pressure, and a fixed observer guard prevents unintended critical changes. Peak heights encode the schedule order; saddle intervals encode the admissible attachments.

Theorem 7.1 (exact physical lift). *For the constructed radius-two interval tree,*

$$\boxed{q\text{-band instance is YES} \iff \text{interval-tree one-polarity observer-safety instance is YES.}}$$

The construction is polynomial and uses rational endpoints.

The exact conjugacy and tree-realization proof are in Appendix E.

7.3 Physical complexity theorem

A YES instance has a polynomial-bit rational realization and polynomial-size elder/tie certificate. On a tree, all maximin attachments are unique-path minima, so verification is polynomial.

Theorem 7.2 (physical observer-safety complexity). *The problem of deciding whether an interval-valued rooted tree and observer x admit one realization and one admissible elder/tie resolution for which the one-polarity EF-FG saliency is H -safe at x is NP-complete even when*

$$\boxed{\theta = \frac{1}{2}, \quad H = 0,}$$

the tree has radius two about the observer, and all interval endpoints are rational.

Thus the hardness does not require cycles, deep tree topology, or variable model parameters. Polynomial witness size and verification are proved in Appendix E.

8 Self-dual ownership complexity

8.1 Dark-anchor neutralization

To lift one-polarity safety to the full self-dual ownership predicate, add one constant-size low-valued leaf d , the **dark anchor**. It is chosen so that it is not a bright primitive maximum and hence leaves every bright observer quantity unchanged.

In the inverted field $-F$, the anchor becomes the essential dark atom. Every other primitive dark birth reaches the observer through a path whose bottleneck equals the observer value. Consequently every finite dark birth has the same observer attachment as the anchor's running elder pressure.

Theorem 8.1 (dark-anchor neutralization). *For every realization and every admissible resolution,*

$$\boxed{\Sigma_{\theta}^{-}(x; F) = 0.}$$

The statement is realization-wise; no independent choice of a dark realization is made. A realization-wise proof is given in Appendix F.

8.2 Possible background ownership

At $\Delta = 0$, after adding the dark anchor,

$$\text{Own}_{0,\theta}(x; F) = \text{background} \iff \Sigma_{\theta}^{+}(x; F) = 0.$$

Thus the one-polarity safe-realization problem is exactly the self-dual possible-background decision problem.

Membership in NP follows from a polynomial-bit scalar realization and polynomial-size resolved elder/tie data for both polarities.

Theorem 8.2 (possible-background ownership complexity). *Possible Background Polarity-Core Ownership is NP-complete even on interval-valued radius-two trees with fixed*

$$\boxed{\theta = \frac{1}{2}, \quad \Delta = 0.}$$

The only additional construction beyond Theorem 7.2 is the constant-size dark anchor. The reduction equivalence and NP-membership argument are in Appendix F.

9 The local/global algorithmic boundary

The complexity theorem is best understood by contrasting it with local witness problems.

9.1 Local possible exceedance

For fixed θ and H , a positive saliency witness can be represented by one candidate birth together with a bounded set of separator/connectivity conditions. On interval-valued cactus graphs those constraints are decidable by polynomial exact routines.

Theorem 9.1 (possible one-polarity exceedance). *On interval-valued cactus graphs,*

$$\exists F, \rho : \Sigma_{\theta}^{\sigma}(x; F, \rho) > H$$

is polynomially decidable for either polarity $\sigma \in \{+, -\}$.

A proof by a finite separator witness and cactus block-cut dynamic program is given in Appendix G.

9.2 Complexity separation

Combining the local witness theorem, the fixed-order collapse, and the hardness results gives:

Theorem 9.2 (EF-FG local/global separation). *For polarity-core saliency:*

1. *possible one-polarity exceedance is polynomially decidable on interval-valued cacti;*
2. *arbitrary fixed-order absorbing-tail reachability is polynomial (quadratic by Theorem 5.1), and the nonincreasing-headroom subclass is linear;*
3. *global possible one-polarity safety is NP-complete on radius-two trees;*
4. *self-dual possible-background ownership is NP-complete on radius-two trees.*

The transition is therefore neither “tree versus cactus” nor merely “deterministic versus uncertain.” It occurs when one realization must coordinate all potential observer records under a global no-danger requirement.

The conclusion follows from Theorem 9.1, Theorem 5.1, and the hardness proofs completed in Appendices D–F.

10 Unbounded common-realization obstruction order

NP-completeness is an algorithmic statement. EF-FG also exhibits an all-orders **structural** local-to-global obstruction.

10.1 Protected record complexes

Let P be a finite family of named strict observer-record obligations. Define

$$K_{\text{raw}}(P) = \{A \subseteq P : \text{one admissible EF-FG realization satisfies every obligation in } A\}.$$

When the native maximal-safe hard-band rule is imposed, define

$$K_{\sigma}(P) = \{A \subseteq P : \text{one admissible canonical realization satisfies every obligation in } A\}.$$

Both are downward-closed simplicial complexes. Their minimal nonfaces measure the order at which local joint feasibility ceases to assemble into one realization.

10.2 Canonical boundary theorem

Fix $r \geq 1$. Start from

$$h_0 = 1, \quad x_0 = 2^r,$$

and take $r + 1$ parameter-identical events

$$(h_i, A_i, B_i) = (0, 2^{r-1}, 1).$$

For a canonical owner use,

$$x' = \max\{1, x/2\}.$$

After k protected uses,

$$x_k = 2^{r-k}, \quad 0 \leq k \leq r.$$

Thus any proper protected subfamily, having size at most r , may be scheduled first and every selected event is a strict canonical record. After r uses, however, $x = 1 = B_i$, so one further strict owner record would require $1 < 1$.

Theorem 10.1 (canonical record boundary). *For every $r \geq 1$ there exists an interval-valued radius-two EF-FG tree with $|P_r| = r + 1$ such that*

$$\boxed{K_\sigma(P_r) = \partial\Delta^r.}$$

This theorem is genuinely section-dependent for the symmetric family: with arbitrary safe attachments, the analogous raw claim can fail already at $r = 2$.

The displayed construction is the full proof: every proper face uses at most r protected events, while a $(r + 1)$ st strict owner use is blocked by $B_i = x = 1$.

10.3 Raw boundary theorem

The section-free construction uses shared maximin geometry. Fix $r \geq 1$ and form a radius-two graph with observer x , gates $g_1, \dots, g_r$, individual peaks $p_1, \dots, p_r$, and one shared peak p_* . Each p_i is connected to x through g_i , while p_* is connected through every gate. Add one fixed essential branch to initialize observer pressure.

Choose

$$F(x) = 5, \quad F(s_0) = 1, \quad F(e) = 8, \quad F(g_i) = u_i \in [2, 3],$$

and allow event peak values in $[6, 7]$. Then

$$\kappa_F(p_i, x) = u_i, \quad \kappa_F(p_*, x) = \max_i u_i.$$

Let

$$P_r = \{p_1^{\text{ob}}, \dots, p_r^{\text{ob}}, p_*^{\text{ob}}\}.$$

Every proper subfamily is realizable. If the shared obligation is absent, choose distinct selected gates and process the selected individual peaks in increasing attachment order. If the shared obligation is present, at least one individual obligation is absent; let an omitted gate supply the maximum value 3, choose all selected individual gate values distinct below 3, process them first, and then process p_* .

The full family is impossible. In every realization, some index j attains the maximum gate value, so

$$\kappa_F(p_*, x) = u_j = \kappa_F(p_j, x).$$

All $r + 1$ events cannot then be strict records in one elder order, because strict records require strictly increasing and hence pairwise distinct attachments.

Theorem 10.2 (raw protected-record boundary). *For every $r \geq 1$,*

$$\boxed{K_{\text{raw}}(P_r) = \partial\Delta^r.}$$

Hence every proper protected observer-record family admits a common scalar realization while the full family does not.

The preceding proper-face construction and duplicate-maximum argument form the full proof.

10.4 Consequence

For every fixed locality order k , choose $r > k$. Every protected subfamily of size at most k is realizable, yet the whole family is not. Therefore

Local-to-global consequence. No bounded order of protected local-feasibility tests certifies global common realizability in general.

The statement is specific to the EF-FG observer-record semantics; it does not claim novelty for simplex boundaries, Helly-type language, or local/global inconsistency in the abstract.

11 Spatial-frame layer and complete EF-FG output

The complexity results above concern the polarity-core ownership projection. EF-FG retains a second, spatial projection because scalar foreground/background structure contains information that observer saliency alone does not encode.

For a fixed realization, competitor extrema determine maximin gates toward the observer. Weak/strong sector incidence, terminal directions, and residual terminal regions produce a spatial label recording which side of a local extremal frame the observer occupies. On trees, much of this structure is rigid; on cactus graphs, multiple route channels require a correlated residual state. The spatial layer is self-dual under $F \mapsto -F$.

The **complete deterministic EF-FG state** is therefore a pair

$$\boxed{(\text{PO}_x(F), \text{SF}_x(F))},$$

accompanied when needed by witness/certificate data. Under interval uncertainty the exact state is not the Cartesian product of marginal possibilities; both projections must be computed from one common realization before projection.

Appendix B gives the exact spatial-frame construction, including pairwise gates, route incidence, terminal thresholds, correlated residual states, and the cactus closure statement. None of those details is used to strengthen the ownership hardness claim; they are retained because they are part of the EF-FG theory rather than an implementation artifact.

12 Reference implementation and executable validation

The software companion has two roles: deterministic reference evaluation and certified testing of theorem instances. It is not used as a substitute for proof.

12.1 Deterministic mode

Given one grayscale field, the implementation computes:

- bright and dark extrema/persistence data;
- observer attachments and record chains;
- bright and dark saliency;
- polarity-core ownership;
- fixed-realization spatial-frame output;
- witness and diagnostic information.

This mode is polynomial because it evaluates one fixed realization; the NP-completeness theorem does not apply to ordinary evaluation of a fixed image.

12.2 Interval-certified mode

For interval-valued inputs, the software distinguishes query classes. Implemented local witness engines are used on the subclasses currently supported by the reference code; the full interval-cactus polynomial theorem is analytic and is not claimed here as a completed software backend. Globally coupled safety/ownership queries are routed to explicit combinatorial search or certificate checking rather than being replaced by marginal envelopes.

12.3 Executable audits

The implementation has been used for regression tests of:

- observer-record death identities;
- grayscale-inversion self-duality;
- ownership partitioning;
- finite bright-record examples;
- deterministic ambiguity;
- weak/strong route incidence and terminal-sector cases;
- interval possible-exceedance examples on paths and cacti;
- explicit small common-realization/order searches;
- the minimal $r = 2$ raw boundary seed.

These tests are reproducibility checks, not premises of the theorems. The paper therefore reports only their role and scope; exhaustive logs remain in the project archive.

13 Discussion

13.1 What the theory adds

EF-FG is not a replacement for merge trees, persistence, mathematical morphology, or uncertain scalar-field topology. It uses those neighboring structures to define a specific foreground/background semantics. Three aspects are central.

First, observer-relative saliency admits a hierarchy-free record normal form. Second, uncertainty is treated by exact common realization rather than by independent coordinate ranges. Third, the same semantics supports both computational and structural local/global separation theorems.

The NP-completeness result and the simplex-boundary theorem are complementary. The former says that choosing one globally safe realization can encode hard combinatorial synchronization. The latter says that even arbitrarily high-order local feasibility can fail to certify global feasibility. Neither theorem is implied by the other.

13.2 Why radius two matters

Hardness on radius-two trees rules out an explanation based on graph-topological complexity. The reduction uses shallow geometry; the combinatorial burden lies in coordinating the scalar realization and event order. Conversely, the raw boundary theorem uses radius-two fan graphs because shared gates are the mechanism that forces the duplicate maximum. The canonical boundary theorem already works on radius-two trees.

13.3 Scope and limitations

Several boundaries should remain explicit.

- The NP-complete ownership predicate is the **polarity-core ownership projection**, not the full spatial-frame mask.
- Deterministic evaluation of one fixed scalar field is not NP-complete.
- The tractable local-exceedance result is proved for the stated structured graph class; it is not a universal polynomial theorem for arbitrary graphs and arbitrary EF-FG predicates.
- The raw boundary theorem is category-specific; it does not assert that every simplicial complex is realizable as an EF-FG obligation complex.
- The present image-analysis experiments are validation and demonstration, not a benchmark claim against learned segmentation systems.

13.4 Open directions

The theorem package suggests several natural problems: characterize which finite complexes occur as EF-FG common-realization complexes; determine parameterized complexity in the number of incomparable records or uncertain gates; extend exact perturbation semantics beyond boxes; develop multichannel and spatiotemporal versions; and determine the complexity of the full spatial-frame projection under uncertainty.

A broader exact-realization viewpoint also motivates comparing EF-FG obligation complexes with protected-realization invariants developed in other domains [15]. The EF-FG theorems proved here, however, are self-contained and require no external realization-theory axiom.

14 Conclusion

EF-FG provides a self-dual mathematical theory in which foreground/background structure is organized by extrema, persistence, maximin observer geometry, and exact common-realization uncertainty. Its two projections are deliberately separate: polarity-core ownership captures persistent observer-relative explanation, while the spatial-frame layer captures carrier and terminal geometry.

For the ownership layer, observer records eliminate the need to carry the full merge hierarchy. This normal form leads to an exact q-band scheduling kernel. Fixed schedules are easy to evaluate, but choosing a globally safe order is NP-complete. The kernel lifts exactly to interval-valued radius-two scalar trees, and a constant-size dark anchor yields NP-completeness of Possible Background Polarity-Core Ownership at fixed $\theta = 1/2$ and $\Delta = 0$.

The same common-realization semantics also supports an all-orders structural obstruction: for every r , explicit EF-FG families realize the simplex boundary $\partial\Delta^r$ as a protected record-feasibility complex, both canonically and, with shared maximin geometry, in a section-free raw form. Hence no bounded local consistency order can certify global common realizability in general.

Together these results identify a precise boundary: **individual danger may be witnessed locally, but global absence of danger and global assembly of protected records can require genuinely global coordination inside one scalar realization.**

A Observer-record proofs and exact common-realization conjugacy

This appendix supplies the proofs used in Sections 3–4.

A.1 Maximin equality

For vertices p, q, x , if $\kappa_F(p, x) > \kappa_F(q, x)$, then

$$\boxed{\kappa_F(p, q) = \kappa_F(q, x).}$$

Indeed, the maximin strong-triangle inequality gives

$$\kappa_F(p, q) \geq \min\{\kappa_F(p, x), \kappa_F(q, x)\} = \kappa_F(q, x).$$

If the inequality were strict, applying the same inequality to $q \rightarrow p \rightarrow x$ would give

$$\kappa_F(q, x) \geq \min\{\kappa_F(q, p), \kappa_F(p, x)\} > \kappa_F(q, x),$$

a contradiction.

A.2 Proof of Theorem 3.1

Assume first that λ is an observer record, so $a_\lambda(x) > a_\mu(x)$ for every elder atom μ . Choose birth-carrier representatives e_λ, e_μ . By A.1,

$$\kappa_F(e_\lambda, e_\mu) = a_\mu(x).$$

The ordinary elder death of λ is the largest merge level at which it meets an elder atom, hence

$$d(\lambda) = \max_{\mu \in O(\lambda)} \kappa_F(e_\lambda, e_\mu) = \max_{\mu \in O(\lambda)} a_\mu(x) = D_x(\lambda).$$

If λ is not a record, choose an elder μ with $a_\mu(x) = D_x(\lambda) \geq a_\lambda(x)$. Then

$$\kappa_F(e_\lambda, e_\mu) \geq \min\{a_\lambda(x), a_\mu(x)\} = a_\lambda(x),$$

so $d(\lambda) \geq a_\lambda(x)$. Its observer-relative depth is therefore nonpositive and it cannot satisfy the strict core inequality for any $\theta > 0$. This proves both record death and nonrecord elimination.

A.3 Adjacent-record chain and supply profile

Order the records $r_1, \dots, r_m$ from elder to younger. Because r_i strictly raises the running attachment, its elder pressure is the previous record value: $D_x(r_i) = a_{i-1}$. By A.2,

$$\boxed{d(r_i) = a_{i-1}.}$$

Thus each finite contribution is determined by (B_i, a_{i-1}, a_i) . The danger conditions are exactly

$$B_i - a_{i-1} > H, \quad a_i > \theta B_i + (1 - \theta)a_{i-1}.$$

Define $E_x(b)$ to be the greatest observer attachment supplied by births older than level b . The function is monotone stepwise; a jump occurs exactly when a younger birth strictly raises the running maximum, i.e. exactly at an observer record. Its left value is D , its right value is A , and A.2 gives persistence $B - D$. Hence the one-profile criterion in Section 3.3 follows by checking the displayed two danger inequalities at every finite jump. On a rooted tree, maximin connectivity to a parent is the minimum of the edge/vertex bottleneck and the child-subtree value, while independent child branches combine by maximum; induction on the rooted tree gives the stated max/min recurrence.

A.4 Self-dual conjugacy

The map $F \mapsto -F$ is a bijection between realizations of $\mathcal{P}$ and $-\mathcal{P}$. Superlevel births of $-F$ are precisely sublevel births of F ; maximin connectivity, elder order, record status, and all strict/equality cases are transported under the polarity relabeling. Therefore the joint tuple

$$(\Sigma_\theta^+, \Sigma_\theta^-, \text{Own}_{\Delta, \theta}, \text{SF}, \text{record data})$$

is conjugated realization by realization. Existential and universal projections of this exact relation are consequently exchanged in the claimed bright/dark fashion. This proves Theorem 4.1.

B Exact spatial-frame construction

The ownership projection does not encode route geometry. This appendix fixes one polarity-normalized realization and gives the second EF-FG projection used in Section 11.

Fix a spatial seed threshold $\Delta_{\text{seed}} \geq 0$. Let S be the finite set of nonessential birth atoms satisfying

$$B_p - d(p) > \Delta_{\text{seed}}.$$

Choose one representative vertex for each $p \in S$. For $v \in V$ define

$$\kappa_p(v) = \kappa_F(p, v), \quad \beta(v) = \max_{r \in S} \kappa_r(v), \quad W(v) = \{p \in S : \kappa_p(v) = \beta(v)\}.$$

The *rooted co-winner basin* $\mathcal{B}_p$ is the connected component containing p in the induced set $\{v : p \in W(v)\}$. The rooted unique-winner basin is defined analogously from $W(v) = \{p\}$. Selected seeds p, q are *carrier-adjacent* iff

$$\mathcal{B}_p \cap \mathcal{B}_q \neq \emptyset.$$

For an adjacent pair put

$$s(p, q) = \max_{v \in \mathcal{B}_p \cap \mathcal{B}_q} \beta(v).$$

The *top gate carrier* is the induced set

$$\{v \in \mathcal{B}_p \cap \mathcal{B}_q : \beta(v) = s(p, q)\},$$

retained componentwise rather than collapsed to one arbitrary point.

Fix an ordered pair (p, q) and $x \in \mathcal{B}_p$. For a top gate component C , a simple path in $\mathcal{B}_p$ from p to a vertex of C is an admissible top route. The component is *weakly incident through* x when at least one such route contains x , and *strongly incident through* x when every such route contains x . Thus strong incidence implies weak incidence; on a cactus the two conditions reduce respectively to route existence and a no-bypass (vertex-dominator) test in the rooted carrier subgraph.

For a bright sector define the midpoint margin

$$\mu_{p,q,x}^+ = 2F(x) - F(p) - s(p, q).$$

The three signs > 0 , $= 0$, and < 0 are retained, and are interpreted together with weak/strong incidence. The dark sector state is the same construction applied to $-F$.

To make boundary information explicit, let $T \subseteq V$ be a designated terminal set supplied with the spatial query. An exposed terminal for seed p is a vertex $t \in T \cap \mathcal{B}_p$. Its gate is

$$s_\partial(p, t) = \kappa_{F|_{\mathcal{B}_p}}(p, t).$$

Put $s = s_\partial(p, t)$ and, for $v \in \mathcal{B}_p$,

$$\alpha(v) = \kappa_{F|\mathcal{B}_p}(p, v), \quad \gamma(v) = \kappa_{F|\mathcal{B}_p}(t, v).$$

Vertices with $\alpha > s$ are forced to the peak side; vertices with $\gamma > s$ are forced to the terminal side; only the tie zone $\alpha = \gamma = s$ can be assigned by residual resolution. A *residual terminal resolution* is a connected terminal-side region R containing t , containing every forced terminal-side vertex in its connected terminal component, excluding forced peak-side vertices, and using only tie-zone vertices for the remaining choice. The intrinsic family of all such regions is denoted $\mathfrak{R}_{p,t}^{\text{res}}$.

For $R \in \mathfrak{R}_{p,t}^{\text{res}}$ define its floor and cap

$$L(R) = \min_{z \in R} F(z), \quad U(R) = \max_{z \in R} F(z).$$

For the canonical terminal side $L(R) = s_\partial(p, t)$. At threshold Δ , define

$$T_\Delta^+(L, U) = \max \left\{ L + \Delta, \frac{L + U}{2} \right\}, \quad T_\Delta^-(L, U) = \min \left\{ U - \Delta, \frac{L + U}{2} \right\}.$$

A query value y is terminal-dark when $y < T_\Delta^-$, neutral when $T_\Delta^- \leq y \leq T_\Delta^+$, and terminal-bright when $y > T_\Delta^+$. Grayscale inversion exchanges the bright and dark inequalities and fixes the neutral band.

Under interval uncertainty, these quantities must remain correlated. The exact residual object is therefore

$$\text{ResRel}_{p,t,x} = \{(s, U, y, \chi) : \text{one admissible realization and one } R \in \mathfrak{R}_{p,t}^{\text{res}} \text{ realize the tuple}\},$$

where $y = F(x)$ and $\chi = \mathbf{1}[x \in R]$. Independent extrema of s, U, y, χ may not be recombined. On cactus graphs the block-cut decomposition gives constant-width route interfaces, so pairwise gates, weak/strong incidence, terminal exposure, residual connectedness, and the correlated floor/cap/query relation admit polynomial exact dynamic programs. This is a spatial-frame statement and is distinct from the polarity-core complexity theorems below.

C q-band derivation and fixed-order theorem

Let $c = 1 - \theta$. In a critical band with floor q , no schedule may lower the running birth cap below q while an event with minimum birth q remains unresolved. If $q \leq D + H$, every remaining birth $B \leq q$ is persistence-safe, so only the hard regime $q > D + H$ matters.

For one hard-band event, put $U_i = \min\{M, u_i\}$. It has a q -preserving safe realization iff

$$\alpha_i \leq \theta U_i + cD.$$

The coordinatewise maximal safe attachment is

$$a_i^* = \min\{\beta_i, \theta U_i + cD\},$$

and every other q -preserving realization is dominated by $(U_i, \max\{D, a_i^*\})$.

Translate by

$$h = M - q, \quad x = q - D,$$

and $h_i = u_i - q$, $A_i = q - \alpha_i$, $B_i = q - \beta_i$. With $s_i = \min\{h, h_i\}$, owner legality becomes

$$B_i < x, \quad cx - \theta s_i \leq A_i,$$

and the transition is

$$(h, x) \mapsto (s_i, \max\{B_i, cx - \theta s_i\}).$$

The absorbing safe tail is $x \leq H$. For a fixed legal owner word $i_1, \dots, i_k$ with realized headrooms s_t , repeated substitution gives

$$x_k = \max \left\{ B_{i_k}, cB_{i_{k-1}} - \theta s_k, \dots, c^k x_0 - \theta \sum_{t=1}^k c^{k-t} s_t \right\}.$$

Thus evaluation of a fixed word is linear-size max-affine arithmetic.

For an arbitrary fixed order, let

$$\mathcal{H} = \{h_0\} \cup \{\min(h_0, h_i) : 1 \leq i \leq n\}.$$

After every prefix, the current headroom belongs to $\mathcal{H}$. For $\eta \in \mathcal{H}$, let $g_j(\eta)$ be the minimum hard deficit $x > H$ reachable after processing the first j events with current headroom exactly η ; use $+\infty$ if no such state exists, and retain one absorbing flag when some subsequence has reached $x \leq H$. Initially $g_0(h_0) = x_0$.

Process event i from every finite state (η, g) in two ways. The skip transition retains (η, g) . For the use transition set

$$s = \min\{\eta, h_i\}.$$

It is legal exactly when

$$B_i < g, \quad cg - \theta s \leq A_i,$$

and then outputs

$$y = \max\{B_i, cg - \theta s\}.$$

If $y \leq H$, set the absorbing flag; otherwise update $g_i(s) \leftarrow \min\{g_i(s), y\}$.

For a fixed headroom η , a smaller hard deficit dominates every larger one for future absorbing-tail reachability. If an event is feasible at the smaller value, monotonicity gives no larger output there. If the smaller value is at or below an event floor B_i , any use from a larger live predecessor outputs at least B_i and cannot beat the smaller state. If the smaller value violates the upper feasibility inequality, every larger value violates it as well. Hence keeping only $g_j(\eta)$ is exact. There are at most $n + 1$ headroom values and n prefixes, so the dynamic program uses $O(n^2)$ event-state transitions. This proves the first part of Theorem 5.1.

Now assume

$$h_0 \geq h_1 \geq \dots \geq h_n.$$

Whenever event i is used, its effective headroom is exactly h_i , independent of earlier skips. Put

$$U_i = \frac{A_i + \theta h_i}{c}, \quad T_i(x) = \max\{B_i, cx - \theta h_i\}.$$

At a hard state event i is usable exactly when $B_i < x \leq U_i$, and then $T_i(x) < x$. Scan left to right and use the event whenever it is usable. If the current greedy value is g , every alternative predecessor is at least g . If i is usable at g , monotonicity of T_i makes $T_i(g)$ the smallest successor. If $g \leq B_i$, every live use from a larger state outputs at least $B_i \geq g$. If $g > U_i$, the event is infeasible from every larger state. Induction proves the prefix-minimum invariant and the $O(n)$ greedy specialization.

D Complete q-band NP-completeness reduction

We now give the reduction underlying Theorem 6.1. Fix $\theta = c = 1/2$ and $H = 0$, so

$$(h, x) \mapsto \left(s_i, \max \left\{ B_i, \frac{x - s_i}{2} \right\} \right), \quad s_i = \min\{h, h_i\},$$

with legality $B_i < x$ and $(x - s_i)/2 \leq A_i$. We reduce from Monotone NAE-3-SAT-E4.

D.1 The normalized Boolean interface

We first give the exact-one control used for a Boolean variable. Work at three consecutive caps

$$h_0, \quad h_1 = h_0 - 1, \quad h_2 = h_0 - 2,$$

and let both marker banks have size $m = 8$. Fix $Z = 1$, take

$$h_2 = 2^{80}, \quad h_0 = h_2 + 2,$$

and set

$$y_0 = 2^{28}(Z + h_0 + 10), \quad X = y_0 - h_0.$$

The module enters at (h_0, X) . Each A -marker has headroom h_0 and each B -marker headroom h_1 ; all markers have the common large attachment threshold

$$A_{\text{big}} = X.$$

Choose their floors in two disjoint ordered intervals far below every assignment-time core value, with every B -marker floor larger than every A -marker floor. Distinct dyadic values in, say, $(-2, -3/2)$ for the A bank and $(-1, -1/2)$ for the B bank suffice for the local construction. Since every intended marker step is then core-limited, the numerical values of these floors do not enter the displayed recurrences. In the final composed reduction they may be translated to a small positive ordered interval, still below all assignment-time core outputs; this preserves every comparison below and permits the exact cleanup used in Section D.4.

At cap h_0 , put $y = x + h_0$. An A -marker acts by $y \mapsto y/2$, so after t such markers

$$x_A(t) = \frac{y_0}{2^t} - h_0.$$

The **early A pivot** is

$$E_A : \quad h = h_1, \quad B = X - 1, \quad A = A_{\text{big}}.$$

At prefix zero it sends (h_0, X) to $(h_1, X - 1)$. After even one A -marker we have $x_A(1) < X - 1$, so E_A is permanently owner-dead. Thus E_A is exactly the retain- A exit.

For the **late A pivot**, define

$$C_A(t) = \frac{x_A(t) - h_1}{2}$$

and choose

$$B_{L_A} = 0, \quad A_{L_A} = \frac{C_A(7) + C_A(8)}{2}.$$

Hence L_A is infeasible before all eight A -markers have been consumed and becomes feasible at $t = 8$. The two first-fork outputs are therefore

$$x_H = X - 1$$

on the retain- A branch and

$$x_L = C_A(8) = \frac{y_0}{2^9} - h_0 + \frac{1}{2}$$

on the consume- A branch.

Now work at cap h_1 . After t B -markers write

$$x_H(t) = \frac{x_H + h_1}{2^t} - h_1, \quad x_L(t) = \frac{x_L + h_1}{2^t} - h_1.$$

The chosen scale gives $x_H(8) > x_L$. We use the sharpened second fork needed later in the global reduction. Set

$$B_{E_B} = x_L - \frac{1}{4}, \quad U_{E_B} = x_H(8) - \frac{1}{4}, \quad A_{E_B} = \frac{U_{E_B} - h_2}{2}.$$

At the low input x_L , the early pivot E_B is live and feasible and is floor-limited, so it sends the branch exactly to

$$\boxed{x_- = x_L - \frac{1}{4}}.$$

After one low-branch B -marker it is owner-dead. On every high-branch prefix $0 \leq t \leq 8$, the scalar is at least $x_H(8) > U_{E_B}$, so the corresponding pivot candidate exceeds A_{E_B} and E_B is infeasible there. Thus the consume- A branch can complete only by retaining the whole B bank through immediate E_B .

For the **late B pivot**, put

$$B_{L_B} = x_L + 1, \quad C_H(t) = \frac{x_H(t) - h_2}{2}, \quad A_{L_B} = \frac{C_H(7) + C_H(8)}{2}.$$

The floor B_{L_B} is above every scalar on the low branch, so L_B is owner-dead there. On the high branch it is infeasible through the first seven B -markers and becomes feasible after the eighth. At that point

$$C_H(8) = x_L + 1 - 2^{-8} < B_{L_B},$$

so it is floor-limited and sends the branch exactly to

$$\boxed{x_+ = x_L + 1}.$$

Consequently the only two local completions to cap h_2 are

$$E_A, B_1, \dots, B_8, L_B$$

and

$$A_1, \dots, A_8, L_A, E_B.$$

The first retains the entire A bank and consumes the entire B bank; the second consumes the entire A bank and retains the entire B bank. There is no completion retaining both banks, consuming both banks, or retaining a proper nonempty subbank. Thus the module is an exact-one Boolean selector.

The two exits share cap h_2 and satisfy

$$x_+ - x_- = \frac{5}{4}.$$

Consume eighteen copies of the explicit plateau normalizer

$$N : \quad h_N = h_2, \quad B_N = 0, \quad A_N = \frac{x_+ - h_2}{2}.$$

At every intended normalizer input $x \leq x_+$, its core candidate is $(x - h_2)/2 \leq A_N$; it is positive throughout these eighteen uses, so the zero floor is inactive. Hence, in shifted coordinate $y = x + h_2$, each copy acts exactly by $y \mapsto y/2$. The normalized pair is

$$S_- = (h_2, \ell), \quad S_+ = (h_2, \ell + \delta),$$

where

$$\delta = \frac{5}{2^{20}}, \quad \ell = h_2 + 26 - \frac{7}{2^{20}}.$$

Finally choose the low-branch adapter cap $C = h_2 - 1$. A low adapter with threshold strictly between the two branch candidates is feasible only on S_- , while a high adapter with floor $\ell + \delta/2$ is owner-dead on S_- and live on S_+ . The resulting transfer-mode entry scalars are

$$a = \frac{27}{2} - \frac{7}{2^{21}}, \quad b = h_2 + 26 - \frac{9}{2^{21}},$$

with $b/a > 2^{75}$. Hence a completed variable selects exactly one Boolean transfer mode and designates exactly one of two eight-token payload banks for deletion, leaving the other for the final clause audits.

D.2 Exact headroom certification

At a transfer state (C, X) let the intended payload have effective headroom $0 < h < C$, every wrong candidate have headroom $t \neq h$, and assume $X > 2h$. A candidate gives $y(t) = (X - t)/2$. After a high-headroom first gate the core candidate is $z(t) = (X - 3t)/4$. Choose

$$\max\{\max_{t>h} y(t), \max_{t\leq h} z(t)\} < Q < y(h).$$

A first gate with floor Q kills every $t > h$ substitute and floor-normalizes all $t \leq h$ survivors to common scalar Q . The target then has next core candidate $c_h = (Q - h)/2$, whereas every $t < h$ gives $c_t = (Q - t)/2 > c_h$. Choose

$$c_h < R < A_2 < \min_{t<h} c_t.$$

A second gate with floor R and threshold A_2 accepts and floor-normalizes only the target. Thus payload identity is certified entirely by headroom; its own floor B remains free for clause logic.

For the eight-step deletion word use, at stage j ,

$$X_j = \frac{S}{16^{j-1}}, \quad h_j = \frac{X_j}{4},$$

$$Q_j = \frac{17}{64}X_j, \quad R_j = \frac{1}{16}X_j, \quad A_j^{(2)} = \frac{3}{32}X_j.$$

For the first gate choose the explicit attachment threshold

$$A_j^{(1)} = X_j.$$

Every intended or competing Gate-1 core candidate considered here is at most $X_j/2 < A_j^{(1)}$, so this threshold never removes an intended Gate-1 use.

The intended branch satisfies

$$\frac{X_j}{128} < \frac{X_j}{16} < \frac{3X_j}{32} < \frac{X_j}{8},$$

so the second gate accepts it and outputs $X_{j+1} = X_j/16$, while higher substitutes are rejected by Gate 1 and lower substitutes by Gate 2. This deletes exactly the designated eight-token bank.

D.3 Clause audit

At cap $h = 0$, an audit token acts by $x \mapsto \max\{B, x/2\}$. For a clause scale H let high tokens have floor H and low tokens floor $L \leq H/2$, and enter at $x_0 = 4H$. Three high tokens fail because

$$4H \rightarrow 2H \rightarrow H$$

and the third high token is owner-dead. If at least one token is low, all three designated uses are possible and the state reaches $H/2$; a progress event at floor/threshold $H/4$ certifies completion. Hence one orientation checks *not all True*. Reversing the high/low truth orientation checks *not all False*. The conjunction is exactly NAE.

Audit stages are chained by

$$H_{s+1} = H_s/16, \quad L_s = H_s/2.$$

After the two audit copies define the final positive audit output

$$\boxed{\varepsilon := H_{2m}/4.}$$

The explicit scale inequalities of the control construction place every intended input and every intended core output of a convenience zero-floor event at or above this final scale, with strict input liveness and attachment allowance $A_i \geq \varepsilon$. The successful word terminates with q-deficit exactly ε .

Earlier-stage tokens are already owner-dead at later stages. Therefore any foreign substitution points strictly forward. Following the missing-token deficit reaches a maximal stage with no later substitute, so no globally successful schedule can have net foreign-token displacement.

D.4 Global scale separation and phase-leak repair

The q-band kernel is positively homogeneous. Put

$$\lambda_i = \sigma^{i-1}, \quad \sigma = 2^{-512},$$

and let $X_0 = X$ denote the normalized variable-input scalar from Section D.1.

Lemma D.1 (scale separation and exact inter-variable composition). *For every variable $i < n$, the two truth branches can be reconverged at one explicit positive raw port and transferred to the input port of variable $i + 1$ by a constant number of q-band events. Moreover, at every certified deletion stage of variable i , payloads belonging to earlier or later variables cannot substitute for the intended payload. All event coordinates are dyadic rationals of polynomial bit length.*

Proof. After the eight certified deletions, the high and low transfer modes have scalars

$$S_H = \frac{b}{16^8}, \quad S_L = \frac{a}{16^8}.$$

Define the normalized raw port

$$C_{\text{raw}} = \frac{S_L}{2}, \quad P_{\text{raw}} = \frac{3S_L}{4}.$$

Use eighty copies of the reconvergence normalizer

$$N_{\text{rec}} : \quad h = C_{\text{raw}}, \quad B = P_{\text{raw}}, \quad A = S_H.$$

On the low branch the first copy is floor-limited and reaches $(C_{\text{raw}}, P_{\text{raw}})$. On the high branch, with $y = x + C_{\text{raw}}$, the recurrence is

$$y \mapsto \max \left\{ P_{\text{raw}} + C_{\text{raw}}, \frac{y}{2} \right\}.$$

Since $b < 2^{81}$, $a > 2^3$, and hence $S_H/S_L = b/a < 2^{78}$, eighty copies are more than enough to reach the same floor. Once the floor is reached, the unused copies are owner-dead. Thus both truth branches reach exactly the same raw port.

Scale variable i by λ_i . From the normalized constants,

$$2^{-31} < C_{\text{raw}} < P_{\text{raw}} < 2^{-28}, \quad h_0 < 2^{81}, \quad 2^{107} < X_0 < 2^{108}, \quad X_0 + h_0 = y_0 > 2^{108}.$$

Hence

$$\sigma h_0 < 2^{-431} < C_{\text{raw}}, \quad \sigma X_0 < 2^{-404} < P_{\text{raw}},$$

so the first tail event is both cap-lowering and owner-live. Append 384 copies of

$$T_i : \quad h_T = \lambda_{i+1} h_0, \quad B_T = \lambda_{i+1} X_0, \quad A_T = \lambda_i P_{\text{raw}}.$$

The first and every later intended use is attachment-feasible because its core candidate is below $\lambda_i P_{\text{raw}}/2 < A_T$. After the first copy the cap is $\lambda_{i+1} h_0$; in the shifted coordinate $y = x + \lambda_{i+1} h_0$ the recurrence is exactly

$$y \mapsto \max \left\{ \lambda_{i+1} (X_0 + h_0), \frac{y}{2} \right\}.$$

At the start of the tail,

$$y < \lambda_i 2^{-27},$$

whereas the target floor satisfies

$$\lambda_{i+1} (X_0 + h_0) > \lambda_i 2^{-404}.$$

Consequently

$$\frac{y}{2^{384}} < \lambda_i 2^{-411} < \lambda_{i+1} (X_0 + h_0),$$

so the target floor is reached by or before the 384th copy. At that moment the state is exactly

$$\lambda_{i+1} (h_0, X_0),$$

and every unused tail copy is owner-dead.

For separation, every transfer headroom of variable i lies in

$$\lambda_i (2^{-30}, 2^{80}).$$

Hence every future-variable transfer headroom is at most

$$2^{80} \lambda_{i+1} = 2^{-432} \lambda_i,$$

whereas every current intended transfer headroom exceeds $2^{-30} \lambda_i$. The ratio is therefore greater than 2^{402} . Future payloads are lower/over-drop candidates and fail Gate 2. A retained payload from the immediately preceding variable already has headroom greater than $2^{-30} \lambda_{i-1} = 2^{482} \lambda_i$, while the current cap is below $h_0 \lambda_i < 2^{81} \lambda_i$. Hence every earlier retained payload appears at the current cap as a higher/no-drop candidate and fails Gate 1. Thus no foreign payload can replace an intended deletion token. Finally, the exponent of λ_i is $512(i-1)$, and all other constants have fixed dyadic encoding, so the complete construction has polynomial bit length. $\square$

The same scale bounds keep future transfer and audit gates attachment-infeasible throughout the preceding handoff: before the next canonical input one has $x > \lambda_{i+1}X_0$, with $X_0 > 2^{107}$, while the normalized transfer/audit headrooms and attachment thresholds are below 2^{81} . Thus their core candidate is greater than $2^{105}\lambda_{i+1}$, far above the corresponding attachment threshold.

It remains to exclude premature use of the high-threshold assignment controls. Choose the residual RN136 marker floors positive and in their required order, but below

$$\varepsilon_{\text{ctl}} = \frac{P_{\text{raw}}}{2^{13}}.$$

At the common raw port use one drop event

$$D : \quad h_D = 0, \quad B_D = 0, \quad A_D = P_{\text{raw}}/2,$$

which sends the canonical scalar to $P_{\text{raw}}/2$. Eight residual controls are then core-limited and yield

$$x_8 = \frac{P_{\text{raw}}}{2^9}.$$

The cleanup gate

$$G_{\text{ctl}} : \quad h = 0, \quad B = 0, \quad A = \frac{P_{\text{raw}}}{2^{10}}$$

is feasible after eight controls and infeasible after only seven. Therefore premature consumption of an assignment marker destroys both future truth branches. Likewise, the branch adapter requires exactly eighteen common post-choice normalizers; using one during the preceding handoff leaves only seventeen after the scale floor erases its scalar effect, so the certified adapter input cannot be reached.

The only remaining high-threshold leak is a premature early pivot E_A . Write its output as

$$x_E = X - 1 + \xi, \quad 0 \leq \xi \leq 1.$$

After it fires, all remaining assignment markers act as $y \mapsto y/2$ in shifted coordinate. No truth exit is possible before eight contractions, and more than eight destroys the cleanup count. After exactly eight contractions, either the low route $L_A \rightarrow E_B$ reaches exactly x_- , or the direct mixed E_B value lies in

$$x_L + 1 - \frac{1}{256} \leq x_{\text{mid}} \leq x_L + 1 - \frac{1}{512}.$$

Since the truth midpoint is $x_L + 3/8$, every mixed state lies strictly on the high side and the branch adapter routes it to the canonical high mode. Thus premature phase entry cannot create a third truth value or retain both payload banks.

D.5 Correctness and NP membership

Given a NAE assignment, choose the corresponding truth branch in each variable, delete the opposite bank by the certified eight-step words, traverse the scale ladder, and run both audit copies. Every clause contains both truth values, so both not-all audits pass.

Conversely, every successful completion induces exactly one retained truth bank per variable. Headroom certification prevents foreign substitution during deletion; phase-leak repair preserves the two-valued interpretation; directed audit self-policing removes net foreign displacement. The first audit copy implies no clause is all True and the second no clause is all False. Hence the induced assignment is NAE-satisfying.

Each variable and clause contributes only constantly many events, so the construction has $O(n + m)$ events. The dyadic scale exponents and all rational coordinates have polynomial encoding length. A YES certificate is the owner-event order. Verification simulates at most N exact rational transitions; at $\theta = 1/2$ denominator exponent grows by at most one per non-floor step. Therefore the problem is in NP. This proves Theorem 6.1.

E Exact physical lift and physical NP membership

The radius-two realization requires event saddles not to exceed the observer. The hardness construction initially uses a few zero floors only for notational convenience, so we first remove them.

Lemma E.1 (positive-floor normalization). *For the q -band hardness family, there is a polynomial-bit rational $\varepsilon > 0$ such that replacing every convenience floor $B_i = 0$ by $B_i = \varepsilon$ preserves YES/NO equivalence. After normalization,*

$$0 < \varepsilon \leq B_i \leq A_i$$

for every event.

Proof. Use the same final audit scale fixed in Appendix D,

$$\varepsilon = H_{2m}/4.$$

It is a positive dyadic rational of polynomial encoding length. The explicit scale inequalities in Appendix D give, for every intended use of a convenience zero-floor event, input $x > \varepsilon$, attachment allowance $A_i \geq \varepsilon$, and intended core output at least ε . Thus the intended completeness schedules are literally unchanged after replacing $B_i = 0$ by $B_i = \varepsilon$; in particular the successful word still ends at q -deficit exactly ε .

For soundness, denote the original and normalized floors by $B_i^{(0)}$ and $B_i^{(\varepsilon)}$, respectively. Thus

$$B_i^{(0)} \leq B_i^{(\varepsilon)},$$

with equality except for convenience zero-floor events, where $B_i^{(0)} = 0$ and $B_i^{(\varepsilon)} = \varepsilon$. Compare any normalized trajectory with the original instance and couple states in the product order

$$h^{(0)} \geq h^{(\varepsilon)}, \quad x^{(0)} \leq x^{(\varepsilon)}.$$

This holds initially. Suppose the normalized schedule next uses event i . If $B_i^{(0)} < x^{(0)}$, then the event is owner-live in the original state; moreover

$$\frac{x^{(0)} - \min\{h^{(0)}, h_i\}}{2} \leq \frac{x^{(\varepsilon)} - \min\{h^{(\varepsilon)}, h_i\}}{2} \leq A_i,$$

so the original use is feasible. Its new deficit is no larger because both its floor and its core candidate are no larger, while its new headroom is no smaller. If instead $x^{(0)} \leq B_i^{(0)}$, then $B_i^{(0)} \leq A_i$ implies $x^{(0)} \leq A_i$, so event i is already passively resolved in the original instance and may be skipped. The product order is preserved in either case. Keeping the larger old headroom only makes every later core-feasibility inequality easier, while the smaller old deficit never worsens terminal safety. Induction therefore turns every normalized successful completion into a successful completion of the original instance. Hence normalization cannot create a false YES. Together with the unchanged intended completeness schedules, YES/NO equivalence is preserved. $\square$

Choose

$$q > \max\{x_0, \max_i A_i\} + 1$$

and truncate h_i to $\bar{h}_i = \min\{h_i, h_0\}$, which changes no reachable effective headroom. Map

$$M = q + h, \quad D = q - x,$$

and

$$\ell_i = q, \quad u_i = q + \bar{h}_i, \quad \alpha_i = q - A_i, \quad \beta_i = q - B_i.$$

By Lemma E.1,

$$0 \leq \alpha_i \leq \beta_i \leq q - \varepsilon < q = \ell_i \leq u_i.$$

Thus the physical birth/attachment constraint is automatic and every event lies in the same critical q -band.

At $(M, D) = (q + h, q - x)$, with $s_i = \min\{h, \bar{h}_i\}$,

$$U_i = \min\{M, u_i\} = q + s_i.$$

The physical core-feasibility inequality

$$\alpha_i \leq \frac{1}{2}U_i + \frac{1}{2}D$$

is exactly $(x - s_i)/2 \leq A_i$, while the event raises the owner exactly when

$$\beta_i > D \iff B_i < x.$$

The owner-changing update translates to

$$x' = \max\left\{B_i, \frac{x - s_i}{2}\right\}, \quad h' = s_i.$$

Hence rectangle-event and q -band owner dynamics are exactly conjugate; owner-dead events correspond to $\beta_i \leq D$ and may be processed passively.

For the tree realization create an observer x with

$$F(x) = q - \varepsilon$$

and, for every event, a branch $x - s_i - p_i$ with

$$F(p_i) \in [q, q + \bar{h}_i], \quad F(s_i) \in [q - A_i, q - B_i].$$

Then

$$F(s_i) \leq q - \varepsilon = F(x) < q \leq F(p_i).$$

Thus p_i is a primitive local maximum, s_i is not, and the unique p_i - x path has minimum exactly $F(s_i)$:

$$\boxed{\kappa_F(p_i, x) = F(s_i).}$$

Different event rectangles are independent because their branches are vertex-disjoint away from x .

Put $M_0 = q + h_0$ and $D_0 = q - x_0$. Add an essential branch $x - s_0 - e$ with $F(e) = M_0$ and $F(s_0) = D_0$, resolving e before equal-height event peaks. Add one guard leaf g adjacent to x with $F(g) = q$ and resolve it after event peaks born at q . Event peak heights realize the schedule order and saddle values realize the attachments. After a prefix $i_1, \dots, i_t$ the observer elder pressure is

$$D_t = \max\{D_0, F(s_{i_1}), \dots, F(s_{i_t})\},$$

exactly the rectangle running supply. By Appendix A, a peak that raises this maximum is an observer record whose death is the previous elder pressure, so its safety inequality is exactly the rectangle/ q -band inequality.

In the successful normalized construction the final q -deficit is ε , hence final elder pressure is $q - \varepsilon = F(x)$. The guard has observer attachment $F(x)$ and is therefore a nonrecord. It creates no positive saliency. Conversely, the guard only adds a constraint, so any observer-safe tree realization already induces a feasible rectangle schedule. Therefore

$$\boxed{q\text{-band YES} \iff \text{radius-two interval-tree observer-safety YES.}}$$

The construction uses $2N + O(1)$ vertices and polynomial rational endpoints.

For NP membership, use as certificate a rational scalar realization, total preorder/tie resolution, bottleneck identities, and record/nonrecord pattern. Once these discrete choices are fixed on a tree, all remaining conditions are rational linear equalities and inequalities: interval bounds, order/tie relations, path-minimum identities, local-maximum conditions, record inequalities, and either $B - D \leq H$ or $a \leq \theta B + (1 - \theta)D$. Replace strict inequalities by one common positive rational slack δ . A nonempty rational polyhedron has a basic feasible rational point of polynomial bit length. The verifier checks all path minima and inequalities exactly in polynomial time. This proves Theorems 7.1–7.2.

F Dark-anchor lift and ownership NP-completeness

Start from the bright hardness tree and attach one fixed sufficiently low leaf d at the observer-side construction so that it is never a bright primitive maximum. Therefore all bright births, attachments, records, and saliencies are unchanged.

In the inverted field $-F$, the leaf d is the essential dark atom. Every finite dark birth reaches the observer through a path whose bottleneck is the observer value, so all finite dark observer attachments equal the running essential dark pressure. Consequently no finite dark atom strictly raises the running elder supply. By the nonrecord elimination theorem in Appendix A,

$$\boxed{\Sigma_{\theta}^{-}(x; F) = 0}$$

for every realization and every admissible tie/elder resolution.

At $\Delta = 0$ we therefore have, realization by realization,

$$\text{Own}_{0,\theta}(x; F) = \text{background} \iff \Sigma_{\theta}^{+}(x; F) = 0.$$

The construction adds only constantly many vertices and leaves the radius-two bound unchanged. Thus the bright one-polarity safe-realization problem reduces exactly to Possible Background Polarity-Core Ownership. NP membership uses the same polynomial-bit realization certificate as Appendix E, evaluated for both polarities. This proves Theorems 8.1–8.2.

G Local possible-exceedance witness on cactus graphs

Fix one polarity and one finite birth atom λ . Write

$$B = B_{\lambda}, \quad a = a_{\lambda}(x), \quad D = D_x(\lambda).$$

By Appendix A, λ contributes more than H iff

$$D < B - H, \quad D < \frac{a - \theta B}{1 - \theta}.$$

Equivalently, there exists a separator scalar ℓ such that

$$D < \ell, \quad B - \ell > H, \quad a > \theta B + (1 - \theta)\ell. \tag{G.1}$$

The last inequality means that a marked representative p and the observer x lie in the same component of

$$\{v : F(v) > t_{\theta}(B, \ell)\}, \quad t_{\theta}(B, \ell) = \theta B + (1 - \theta)\ell.$$

The inequality $D < \ell$ means that the x -component of $\{F \geq \ell\}$ contains no elder capable of suppressing the marked birth. Ties require only two modes: either a strict-higher elder exists,

or one marked disconnected equal- B plateau is resolved elder. Thus a witness uses a constant number of marked vertices and the two retained parameters (B, ℓ) .

Root the cactus block-cut tree at the block containing x . For each threshold graph in the certificate, a block interface stores only:

- (i) whether each articulation port is active;
- (ii) the connectivity partition of active ports;
- (iii) whether the active component is connected to the marked root;
- (iv) constant-many flags for the required presence/absence of an elder/high vertex and for the equal-height tie mode.

The number of thresholds and marks is constant, so every interface has constant state width. A bridge has a constant transition table. A simple cycle is scanned as a two-port finite automaton; child-block messages combine without increasing the state width.

For a vertex interval $[l_v, u_v]$, local modes are determined by comparisons with the affine forms

$$B, \quad \ell, \quad \theta B + (1 - \theta)\ell,$$

together with the fixed offset $B - H$. Their truth pattern changes only when one of these forms meets l_v or u_v , or on the relevant equality faces. Hence $O(n)$ rational lines in the two-dimensional (B, ℓ) plane induce an arrangement with $O(n^2)$ cells and faces. On each cell every vertex has a fixed finite local-mode set, so one bottom-up block-cut DP decides feasibility in $O(n)$ state transitions.

Enumerate the marked birth representative p and, in equal-elder mode, one elder mark q . There are $O(n^2)$ marked choices. For each, inspect the $O(n^2)$ arrangement cells/faces and run the $O(n)$ cactus DP. A straightforward implementation therefore uses

$$\boxed{O(n^5)}$$

rational arithmetic/state-transition operations; no optimality of this bound is claimed. Coefficients have polynomial bit length, so rational cell representatives and comparisons have polynomial bit complexity.

Soundness. An accepted state supplies an actual birth plateau, an admissible strict/equal elder mode, and a separator ℓ satisfying (G.1), so the marked atom contributes saliency $> H$.

Completeness. If one realization has $\Sigma_\theta(x) > H$, choose an atom attaining the contribution, a representative of its birth plateau, and a separator ℓ strictly between D and the two upper bounds in (G.1). That realization lies in one enumerated arrangement cell/face and induces an accepted block-cut state.

This proves Theorem 9.1. Applying the same construction to $-F$ gives the opposite polarity. The local/global separation theorem then follows from Theorem 5.1 and Appendices D–F.

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in the manuscript. A reference implementation and executable regression checks are auxiliary validation tools and are not premises of the proofs. No external dataset is required to reproduce the theoretical results.

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